Somantha Manuranga



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Professional Summary

AI engineer and M.Sc. student in Artificial Intelligence for Industrial Applications with a B.Sc. in Mechanical Engineering. Working student in applied AI at syskomp gehmeyr GmbH, with earlier industry experience in aviation design and production automation. Focus on computer vision, LLMs/NLP and machine learning for industrial processes. Strong in Python and PyTorch, with an emphasis on rigorous evaluation and models that can be deployed (ONNX export, edge-ready rule models, small GPUs).

Work Experience

syskomp gehmeyr GmbH — Amberg, Germany
Working Student (Werkstudent), part-time — *10/2026 – present*
- Apply AI methods to engineering and industrial workflows in a mechanical engineering environment.
- Build and evaluate Python prototypes with an emphasis on traceable, validated results.
- Work with engineering colleagues to turn requirements into practical AI tooling.

KNSI Aviation Engineering — Negombo, Sri Lanka
Design Engineer, full-time — *08/2023 – 09/2024*
- Managed compliance data workflows and technical documentation with SQL and Excel to aviation standards.
- Built automated SQL/Excel reporting templates that cut weekly reporting time by 30 %.
- Analysed production data to optimise workflows, raising process efficiency by 40 %.
- Led the digitalisation of legacy documentation for faster cross-department data access.

CBL Group (Ceylon Biscuits Limited) — Pannipitiya, Sri Lanka
Internship Trainee, Mechatronics & Data — *02/2022 – 11/2022*
- Contributed to process optimisation through data-driven solutions and automation projects.
- Supported end-of-line automation with articulated and parallel robots.
- Designed and prototyped 3D-printed automation components, reducing external sourcing costs and lead times.
- Deployed IoT sensors for real-time monitoring of production assets as groundwork for predictive maintenance.

Education

OTH Amberg-Weiden (Technical University of Applied Sciences) — Amberg, Germany
M.Sc. Artificial Intelligence for Industrial Applications — *03/2025 – present (expected 2027)*
- Modules: Computer Vision, Deep Vision, Deep Learning, NLP & Information Retrieval, Autonomous Robotics, AI Security & Privacy, Machine Learning, Industry 4.0.

Preparation for Master's studies in Germany — Sri Lanka / Germany
Visa process, relocation and German language course — *10/2024 – 02/2025*

University of Sri Jayewardenepura — Sri Lanka
B.Sc. (Hons) Mechanical Engineering, Minor in Mechatronics (GPA 3.27 / 4.0) — *11/2019 – 05/2023*
- Thesis: CNN surrogate model that predicts the aerodynamic drag of 2D shapes from images, replacing CFD runs.
- Project: design and build of a CoreXY 3D printer integrating mechanical design and machine learning.

Technical Skills

Programming & Tools: Python (advanced), C++, SQL, Git/GitHub, REST APIs, pytest, Linux/Bash, Jupyter, LaTeX

ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, CNN, RNN

Computer Vision: OpenCV, YOLOv8, object detection, semantic segmentation (U-Net, DeepLabV3+), Vision Transformers, Grad-CAM, ONNX

NLP & Generative AI: Hugging Face Transformers, LLMs (Qwen, Llama, GPT), fine-tuning (LoRA/PEFT), RAG, LangChain, agentic workflows, SciBERT, LayoutLMv3, Tesseract OCR

Robotics & Industry 4.0: Webots, SLAM, path planning (A*), PID control, MQTT, digital twins, IoT

Data & Backend: pandas, NumPy, Matplotlib, Seaborn, MongoDB, Power BI, ETL pipelines, Flask, Weights & Biases

Mechanical Engineering: CAD/CAM, SolidWorks, CFD simulation, 3D printing, mechatronics

Projects

Industrial AI & Computer Vision

X-ray Casting Defect Detection | *YOLOv8, OpenCV, ONNX* — *2026*
- Fine-tuned YOLOv8n to detect pores, inclusions and shrinkage cavities in GDXray X-ray images of aluminium castings with a series-level split: test mAP50 0.545, precision 0.748.
- Failure analysis by defect size and a CLAHE ablation; ONNX export with identical detections at 60 ms per image on CPU.

Corrosion Segmentation: U-Net vs DeepLabV3+ | *PyTorch, segmentation-models-pytorch* *2026*
- Pixel-level segmentation of four corrosion condition states on steel bridge inspection photos: mean IoU 0.523; U-Net is clearly better on the safety-critical Severe class (IoU 0.325 vs 0.266).

CNN Surrogate Model for Aerodynamic Drag (B.Sc. thesis) | *TensorFlow, Keras, CFD* *2023*
- CNN that predicts the drag coefficient of 2D shapes from images in milliseconds instead of a CFD run; trained on 12,000 simulated shapes after comparing 7 architectures (validation MSE 0.0017).
- 0.06–2.77 % relative error on unseen real shapes, including NACA airfoils, against SolidWorks Flow Simulation.

Industrial Production Analytics Dashboard | *Power BI, DAX, Power Query, SQL* *2025*
- Power BI dashboards on star-schema models with automated ETL pipelines to track manufacturing KPIs.

Stacking Ensemble & Edge-Ready Rule Model | *XGBoost, LightGBM, CatBoost, Optuna* *2025/26*
- Optuna-tuned stacking ensemble with R^2 0.848 on a 273-feature regression task; derived three if-else rules (R^2 0.915) that run on an edge device without ML libraries.

Pneumonia Detection: CNN vs ViT vs Hybrid | *PyTorch, timm, Grad-CAM* *2026*
- Compared ResNet-50, ViT-Base and TinyViT under identical conditions (5-fold CV ensembles): TinyViT reached 94.9 % test accuracy (F1 0.948); assessed how faithful the Grad-CAM explanations are.

Digital IR Thermometer Simulation | *Python, NumPy, signal processing* *2025*
- Physics-based sensor simulation (noise, drift, emissivity bias, spikes) with filtering and outlier rejection.

LLMs, NLP & Document AI

Removing Jailbreak Backdoors from LLMs | *PyTorch, Hugging Face, LoRA/PEFT* *2026*
- Removed a known backdoor from Qwen2.5-1.5B with four-group hybrid supervised fine-tuning on a single 8 GB GPU: attack success rate reduced from 95 % to 1.7 % at 84 % benign accuracy (6.4 % out of distribution).
- Mechanistic analysis with linear probes shows the trigger is still detected (about 97 %): the defence changes the reaction, it does not erase the trigger.

QuantumCircuit Dataset Builder | *SciBERT, Transformers, BeautifulSoup, PyMuPDF* *2025/26*
- Automated pipeline that mines arXiv papers for quantum-circuit figures; fine-tuned SciBERT caption classifier (F1 0.954) producing 250 metadata records and 160 extracted images for image-to-text training.

LayoutLMv3 Form Understanding | *LayoutLMv3, Hugging Face, Tesseract OCR* *2026*
- Fine-tuned LayoutLMv3 for key-value extraction from scanned forms (FUNSD): entity-level F1 0.896.

Robotics & Industry 4.0

Autonomous ROSbot Maze Navigation | *Webots, SLAM, A*, PID, OpenCV* *2026*
- ROSbot that explores unknown mazes with LiDAR and depth-camera mapping, frontier exploration, A* planning and PID control; completed 4 of 5 mazes end to end, fastest full run 54 seconds.

ALIA-Delivery: Hospital Robot Digital Twin | *Webots, MQTT, Flask, A** *2025*
- Digital twin of a hospital floor (team project): a TurtleBot3 receives delivery orders from a Flask control centre over MQTT, plans A* routes and drives to the patient's room.

CoreXY 3D Printer | *mechatronics, CAD/CAM, firmware* *B.Sc. project*
- Designed and built a CoreXY 3D printer from scratch: mechanical design, kinematics and firmware.

Data Analysis

Study Program Intelligence | *Python, pandas, Monte Carlo, MCDA* *2026*
- Reproducible analysis of which new degree programme would attract the most students to OTH Amberg-Weiden: multi-criteria decision analysis under uncertainty with 10,000 Monte Carlo draws.

Students' Social Media Addiction: EDA | *pandas, Seaborn, Matplotlib* *2025*
- Exploratory analysis of 705 students from 110 countries; daily usage correlates with sleep at $r = -0.79$.

Certifications & Training

09/2025: Bavarian-Czech AI Summer School, "Niryo in Action: Controlling Robots with LLMs" (32 hours), Speinshart, Germany

DeepLearning.AI: Deep Learning Specialization; Natural Language Processing Specialization (4 courses); AI for Medicine Specialization (3 courses)

Talk: "Adaptive-RAG" (NAACL 2024 paper): 25-minute scientific talk, AI Conference module, OTH Amberg-Weiden, summer 2026

Languages

English: C1 (professional working level) • **German:** B1 (actively improving) • **Sinhala:** native language

Leadership & Activities

Entrepreneurship: VYROBA TECHS: built a CoreXY 3D printer and ran a small custom-parts business during the pandemic

Football: player at FC Amberg; former captain of the faculty and university teams; University Colours Award for sports excellence (2022)